

Zahir Nikraftar

London, UK | 07424978060 | nikraftarz@gmail.com | [linkedin.com/in/zahir-nikraftar-088372b8](https://www.linkedin.com/in/zahir-nikraftar-088372b8) | [zahir-portfolio-psi.vercel.app](#) | github.com/zahirnik

SUMMARY

AI Engineer and Data Scientist with a Computer Science PhD from Queen Mary University of London (Russell Group, Google DeepMind funded), 7 years in AI and ML — including 2 years specialising in Generative AI. Strong background in large-scale data handling, statistical analysis, feature engineering, predictive modelling, NLP, and computer vision. I build production-grade AI systems end-to-end — RAG pipelines with hybrid retrieval and citation-grounded outputs, tool-calling and multi-agent orchestration, and FastAPI backends and data pipelines optimised for latency, reliability, and real-world performance. Hands-on experience training and serving deep learning models (Transformers, Graph Neural Networks, CNNs), including financial decision-support and trading-signal systems.

TECHNICAL SKILLS

Generative AI, Agents & LLMs: LLM-powered applications, Agentic AI, Tool-Calling Agents, Multi-Agent Orchestration, LangChain, LangGraph, LangSmith, OpenAI API, Hugging Face, Prompt Engineering, LLM Function Calling, Retrieval-Augmented Generation (RAG), Hybrid Retrieval, BM25, Reciprocal Rank Fusion, Reranking.

RAG, Retrieval & Knowledge Graphs: Dense Vector Search, Citation-Grounded Retrieval, Embedding Generation, Metadata Filtering, ChromaDB, Qdrant, pgvector, FAISS, Neo4j, Cypher, Entity Resolution, Subgraph Retrieval.

Backend, Python & Engineering: Python, FastAPI, OpenAPI, REST APIs, Microservices, Celery, Redis, Git, Clean Architecture.

Cloud, Deployment & Observability: AWS (EC2, Lambda, S3, API Gateway), Uvicorn, OpenTelemetry, LangSmith Tracing, Multi-Tenant Architecture, JWT.

Evaluation, Monitoring & Responsible AI: LLM Evaluation, LLM-as-a-Judge, Golden Test Sets, Retrieval Quality Benchmarking, Faithfulness and Grounding Checks, Hallucination Detection, Regression Testing, Guardrails, Human-in-the-Loop Review, Explainable AI, Model Interpretability, Bias Detection.

Data Science & Machine Learning: Supervised & Unsupervised Learning, Classification, Regression, Clustering, Predictive Modelling, Risk Scoring, Propensity & Churn Modelling, Sentiment Analysis, Feature Engineering, Feature Selection, Dimensionality Reduction, Model Validation, Cross-Validation, Hyperparameter Tuning, Time Series Analysis, Statistical Analysis, Experimental Design, scikit-learn, XGBoost, LightGBM, Model Performance Monitoring.

Deep Learning & NLP: PyTorch, TensorFlow, Transformers, Hugging Face, CNNs, RNNs, Graph Neural Networks (GAT, R-GAT, GraphSAGE), Temporal Graph Transformers, Informer / long-sequence models, Diffusion Models, Fine-Tuning, Distributed Multi-GPU Training (SLURM), Computer Vision, Object Detection, Image Classification, Text Classification.

Big Data & Data Engineering: Large-Scale Data Handling, Data Cleaning, Pandas, NumPy, SQL on Large Datasets, PostgreSQL, data pipelines, async job processing, Kafka.

Domain Fit: Financial document intelligence, compliance automation, risk analysis, regulated workflows, behavioural data analysis, customer intelligence, stakeholder-facing analytics.

Frontend (Working Knowledge): React, TypeScript, Vite, Chart.js — self-built functional UIs and dashboards to demo and visualise AI systems; supporting skill rather than a primary specialism.

EXPERIENCE

Convai	London
Lead & Founding Generative AI Engineer	09/2025 – Present

- Designed an agentic orchestration layer (tool-calling agents, multi-turn memory, structured reasoning) with a portable Knowledge Graph schema (Neo4j, Cypher, idempotent upserts, full provenance) for autonomous KG navigation, search, and citation-grounded outputs; deployed an end-to-end RAG + Knowledge Graph platform on dual AWS Lambda/API Gateway and EC2/Uvicorn, processing 1,000+ documents across 8 formats with automated OCR, chunking, embedding, and metadata enrichment.
- Engineered hybrid retrieval (dense vector search via Chroma, Qdrant, pgvector with BM25, Reciprocal Rank Fusion, reranking) plus entity resolution and subgraph retrieval, achieving 25% recall improvement over dense-only baselines and reducing hallucination via auditable source attribution and grounding checks.
- Built feature pipelines and AI-ready dataset workflows including text extraction, cleaning, chunking strategies, and metadata enrichment across heterogeneous formats; designed golden test sets and LLM-as-a-judge evaluation harnesses for retrieval quality, faithfulness, and tool-use accuracy.

- Built a production FastAPI backend (OpenAPI contracts, JWT/API-key auth, route-level RBAC, multi-tenant isolation, Celery async jobs) for 5+ tenants with retry, progress tracking, and audit logging; established observability (JSON logging, OpenTelemetry, LangSmith, Prometheus) and Docker Compose, cutting incident response by 40%.

Permia Sensing

London

Lead AI Developer

05/2024 – Present

- Owned the end-to-end ML lifecycle across large imaging datasets: data ingestion, cleaning, exploratory analysis, labelling strategy, feature engineering, training, validation, deployment, and monitoring.
- Delivered three production deep learning models (object detection, classification, predictive scoring) at 92%+ accuracy with PyTorch training pipelines, hyperparameter tuning, and inference workflows processing 1,000+ images per batch.
- Designed and delivered an end-to-end automation pipeline where a GenAI layer preprocesses incoming data into model-ready inputs, custom local PyTorch deep learning models run inference, and a downstream GenAI layer turns model outputs into structured analysis and client-facing reports, automatically updating the customer dashboard.

University of Tehran

Tehran, Iran

Research and Development

01/2020 – 12/2022

- Built end-to-end ML solutions in Python, scikit-learn, and PyTorch on large-scale datasets, applying statistical analysis, feature engineering, and predictive modelling for risk assessment, scenario evaluation, and decision-support delivered to government and international development partners.
- Designed data cleaning, transformation, and validation pipelines for heterogeneous structured and unstructured data; produced interpretable models and stakeholder-facing analytics translating complex data into operational insight.

SELECTED PROJECTS

GateKeeper AI

Event-Driven Multi-Agent Regulatory Compliance Platform

Event-driven microservices platform (FastAPI, Kafka, multi-tenant PostgreSQL) for regulatory compliance, with a dedicated LLM agent runtime handling semantic search, knowledge-graph navigation, and automated rule extraction from regulatory documents. Backend engineered for security and resilience across 6 services: per-tenant schema isolation, JWT/RBAC, AES-256-GCM field encryption, a SHA-256 tamper-proof audit chain, and circuit-breaker, retry, and bulkhead patterns. Configurable domain packs for financial services, healthcare, and construction.

Contract Review Agent

Agentic RAG for Legal Document Analysis

Agentic RAG service that ingests contract PDFs and reasons about them. A retrieval pipeline (parsing, chunking, embeddings, ChromaDB) feeds an LLM agent with tool-calling and multi-turn memory, while a Neo4j knowledge graph stores market-standard clause benchmarks. The backend performs clause classification (21+ types), risk scoring, missing-clause detection, and grounded semantic search, with switchable OpenAI/Anthropic providers and PDF report generation.

CareFlow AI

Healthcare Workflow Automation Platform

Microservices backend (FastAPI, Celery, PostgreSQL, Redis, S3) automating clinical documentation and compliance for home-care agencies. An AI engine pairs LLMs with speech-to-text (Whisper/Deepgram) to convert caregiver voice notes into structured clinical records, with async Celery workers for transcription and note generation, a state-specific compliance rule engine, and FHIR-based integration for hospital-to-home handoffs.

EDUCATION

PhD in Computer Science

09/2023 – 09/2026

Queen Mary University of London, London, UK — Fully funded scholarship by Google DeepMind

- Designed advanced deep learning architectures (large-scale CNNs, Transformers, Diffusion models) for high-dimensional prediction problems across domains, focused on scalable, interpretable AI and explainable model behaviour.
- Trained large-scale models on HPC clusters (multi-GPU, SLURM), building end-to-end workflows for preprocessing, distributed training, hyperparameter tuning, statistical evaluation, and reproducible experimentation.

INTERESTS

- Data science and applied ML: predictive modelling, feature engineering, behavioural data analysis, and AI-ready dataset design across structured and unstructured sources.

- Deep learning research: novel architectures (CNNs, Transformers, Graph Neural Networks, Diffusion models), distributed training, high-dimensional prediction, and interpretable AI.
- Agentic AI and RAG: tool-calling agents, multi-agent orchestration, hybrid retrieval, Knowledge Graph augmented retrieval, and citation-grounded generation.
- LLM evaluation and responsible AI: retrieval quality, faithfulness, tool-use reliability, guardrails, explainability, and safety-aware GenAI deployment.
- End-to-end ML engineering, production ML, MLOps, scalable APIs, and model serving.